CSE 4109: Assignment

Objective

To connect the concepts learned in this course with *current research developments* in the field.

Instructions

1. **Select a Journal Paper**
   - Choose **one** peer-reviewed research paper published within the last 10 years.
   - The paper must be related to **your group's assigned topic**.
   - Recommended sources: IEEE, ACM, AAAI, IJCAI, NeurIPS, JAIR, Springer, Elsevier.
2. **Read and Understand the Paper**
   - Focus on the **main idea**, not every mathematical detail.
   - Identify the **problem**, **method**, and **contribution**.
3. **Explain in Your Own Words**
   In your report, clearly describe:
   - What problem the paper is trying to solve
   - How the authors approached or solved the problem
   - What results or conclusions they reached
   - No copy-paste. Use **simple, clear language**.
4. **Connect with Course Topics**
   Compare the paper with what you studied in class.
   For example:
   - How does their search algorithm relate to BFS, DFS, A*, Hill-Climbing, etc.?
   - How does their reasoning method relate to Propositional Logic, FOL, Forward/Backward Chaining, Resolution?

Clearly state **similarities and differences.**

5. **Explain the Improvement**
   Describe **how the paper advances or improves** on the classical method from the course.
   Examples:
   - Lower memory usage
   - Faster convergence
   - Better heuristic accuracy
   - Works in real-world scenarios
6. **Your Suggestion / Critique**
   Suggest **how the approach could be improved further**. This can be creative — no need to prove or implement it.
   For example:
   - Use better heuristic
   - Reduce computation time
   - Handle larger input sizes
   - Apply to a different domain

Topic List

1. Modern Heuristic Search Techniques
Study recent advancements in heuristic search that improve on classical A* by reducing memory usage or improving heuristic accuracy, such as pattern databases and memory-bounded search.

2. Monte Carlo Tree Search and Game-Playing AI
Examine how Monte Carlo Tree Search explores large game decision spaces and how systems like AlphaZero combine search with learned evaluation functions.

3. Knowledge Representation in Real-World Systems
Explore how structured knowledge (e.g., semantic networks, rule bases, ontologies) supports reasoning and decision-making in applied domains such as healthcare or intelligent tutoring.

4. Logical Inference and Automated Theorem Proving
Investigate automated reasoning methods like resolution and SAT/SMT solving, and how they are used in areas such as program verification or symbolic planning.

5. Problem Formulation and Representation
Analyze how the way a problem is represented (choice of states, actions, goals) can dramatically change the efficiency or feasibility of solving it.

6. Learning Heuristics Automatically
Study approaches where heuristics are generated or optimized automatically, often using machine learning techniques rather than being designed manually.

7. Planning Under Real-World Constraints
Compare classical deterministic planning models with practical planning scenarios where uncertainty, time, and changing conditions must be handled.

8. Multi-Agent Search and Adversarial Environments
Examine search and decision-making when multiple intelligent agents interact, whether competing (games) or cooperating (robot teams).

9. Constraint Satisfaction Problems (CSPs)
Study how complex constraint-based problems such as scheduling or resource allocation are solved using techniques like backtracking, arc consistency, and heuristic ordering.

Your Topic: (Your Roll%9) +1